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<https://doi.org/10.1057/s41599-026-06673-7>

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Smart city strategy, China's urban innovation and policy effectiveness

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Smart city strategies are the key policy in China's modernization. Yet how such strategies influence urban innovation remains insufficiently understood. This study employs a multi-period difference-in-differences approach within the Technology-Organization-Environment (TOE) framework to analyze panel data from 283 prefecture-level cities in China from 2006 to 2021. It seeks to evaluate the impact of smart city pilot policies on urban innovation levels and their mechanisms of action. The findings suggest that these policies are associated with higher urban innovation, with particularly pronounced effects in central and western regions and in small and medium-sized cities. The mediation effect model shows that improvements in information and communication technology, transportation infrastructure, digital government and social capital are potential channels linking smart city development to higher innovation, and that these mechanisms differ across regions. Spatial analysis further indicates the spillover effects of smart city policies, whereby innovation activities in pilot cities positively affect neighboring cities through regional networks. Overall, the research provides causal evidence and an analytical framework for evaluating place-based innovation policies and their effects on regional development disparities.

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Introduction

In recent years, with the rapid advancement of information technology, the concept of smart cities has emerged, becoming a significant direction for global urban development (Javed et al., 2022). By integrating advanced information and communication technologies (ICT), smart cities can enhance urban management and service efficiency, thereby promoting sustainable urban development (Sorri et al., 2024; K.-L. Wang et al., 2025). Globally, numerous countries and regions have initiated smart city projects to improve the urban quality of life, optimize resource allocation, and stimulate economic growth through technological innovation.

As a novel urban development model, a smart city approach not only aims to improve citizens' quality of life but can also enhance a city's innovation capacity and competitiveness. Therefore, many countries regard fostering innovation and entrepreneurship as a central goal of smart city construction (Giffinger et al., 2007). European initiatives, exemplified by Finland and Italy, make use of open urban data platforms to encourage collaboration among citizens, firms and research institutions in addressing urban challenges, thereby generating a wide range of data-driven innovation projects. This bottom-up, open and collaborative model of digital innovation not only enhances the efficiency of urban service delivery, but also supports the growth of start-ups developing applications such as transport navigation and environmental monitoring (Jaakola et al., 2015; Marchesani & Ceci, 2025). By contrast, Singapore's Smart Nation initiative is state-led and has established a range of grant and tax incentive schemes to support frontier technologies such as artificial intelligence and big data, while the National Research Foundation (NRF) provides ongoing investment to strengthen research and its translation into industry (D. Das & Kwek, 2024). At the same time, smart-city projects in the Middle East, such as Masdar City, have emerged as capital-intensive test beds for clean technologies and as flagship projects for projecting a global image of urban innovation (Cugurullo, 2016; Randeree & Ahmed, 2019). Against this diverse global backdrop, China's strategy is distinct: it is neither purely bottom-up nor confined to a few flagship projects. Instead, China employs a massive, centrally coordinated "pilot" system—a quasi-natural experiment designed to test and diffuse digital governance across hundreds of existing cities simultaneously. Since 2012, the central government has elevated this to a national strategic priority, launching three successive batches of pilot projects covering nearly 300 cities (C. Guo et al., 2024; Yan et al., 2023). Unlike the organic evolution of smart cities in some Western contexts, China's approach represents a form of institutional change, where central authorities leverage pilot status to induce local innovation and industrial upgrading (W. Wang et al., 2025; F. Wu, 2018). By mandating the integration of digital technologies into urban management, these pilots aim to create an innovation-friendly environment and support the shift from factor-driven to innovation-driven growth (Zhu et al., 2024).

Although case studies generally acknowledge the positive link between smart city development and urban innovation, empirical findings remain mixed. Caragliu and Del Bo (2019) report that European cities with smart-city policies generate more patents than their counterparts without such policies. However, other scholars argue that certain smart-city projects have failed to deliver meaningful innovation or inclusive change, and their outcomes vary substantially across cities and contexts (Echebarria et al., 2021; Hollands, 2015). Consequently, key research gaps persist. First, do smart city strategies genuinely catalyze urban innovation, or do they merely create sophisticated digital infrastructures without meaningful innovative capacity? Second, the concrete mechanisms through which smart-city pilots translate

into tangible innovation remain under-explored. Third, the heterogeneity of policy effects across diverse regions and city types is poorly understood; China's vast territory encompasses pronounced disparities in economic development, scientific capacity, and institutional quality that may condition the effectiveness of national smart-city initiatives (Xu et al., 2022; Q. Yang & Elbakri, 2024). Fourth, potential spatial spillover effects of smart-city pilots on the innovation performance of neighboring non-pilot cities warrant systematic investigation.

The rest of this paper is organized as follows: Section "Literature review" provides a systematic review of the foundational literature and pertinent theories concerning smart cities and urban innovation. Section "Hypotheses, Models and Variables" discusses the development of the research hypotheses, the construction of the analytical model, and the selection of the variables. Section "Empirical Results" first presents the empirical findings on the influence of smart cities on urban innovation and then delves into the underlying mechanisms and spatial implications. The discussion section summarizes the main findings and offers policy insights based on the study's results. The final section concludes.

Literature review

Smart-city governance and China's pilot model. Smart cities have gradually evolved from a predominantly technology-centred imaginary into a widely diffused governance paradigm for addressing complex urban challenges, but the institutional configurations through which this paradigm is implemented differ markedly across places (Mora et al., 2023a). Recent comparative studies emphasise the importance of institutional embeddedness, arguing that the success of smart cities depends less on the mere deployment of digital technologies than on how these technologies are articulated with local governance structures, human capital and social institutions (Kummitha & Crutzen, 2019; Marchesani et al., 2023). Kummitha (Kummitha, 2018a) analyses entrepreneurial, corporate-led smart-city visions and warns that treating digital technology as a "technological panacea" raises fundamental concerns about technology appropriation, social polarisation, switching costs and data management. Similarly, Marchesani and Ceci (Marchesani & Ceci, 2025) show that in many European cases, the value of digitalisation is unlocked only when combined with collaborative governance in a "quadruple helix" involving government, industry, academia and citizens. Taken together, these international perspectives suggest that smart-city policies are not merely technical upgrades but complex social innovations that reconfigure urban governance arrangements.

Departing from the more commonly decentralised, market- or community-driven governance models highlighted in this literature, China's smart-city trajectory has been structured around a centrally steered logic of top-level design plus pilot experimentation. The policy sequence began with the Ministry of Housing and Urban–Rural Development's (MOHURD) National Smart City Pilot Work Notice in 2012, which authorised the first batch of 90 pilot cities and framed smart-city construction as a means of integrating digital technologies into urban planning, infrastructure and public-service delivery (K.-L. Wang et al., 2025). Subsequent documents—most notably the National New-type Urbanization Plan (2014–2020) and a series of guidelines on smart-city standard systems—embedded these pilots within a broader agenda of coordinated, people-centred urbanisation and codified technical architectures, data-sharing protocols and evaluation procedures, thereby institutionalising smart-city development as part of China's national digital-governance and

territorial management regime (Yin & Anantsuksomsri, 2022; Yuan et al., 2020).

The designation of smart-city pilots was the outcome of a strategic selection process rather than random assignment. Official documents and empirical analyses indicate that MOHURD prioritised cities with strong governance and high demonstration potential when approving pilot applications; early cohorts were therefore disproportionately composed of provincial capitals, sub-provincial cities and economically advanced prefectures, with later batches gradually extending to smaller and less-developed jurisdictions (C. Guo et al., 2024; Yan et al., 2023). This stratified selection of pilots distinguishes pilot from non-pilot jurisdictions and grants the former selective policy preferences, additional resources and greater latitude for regulatory experimentation, thereby endowing pilot cities with specific political status and institutional advantages (Yi & Shi, 2025). Within this state-led experimental framework, pilot designation forms part of an “experimentation under hierarchy” cycle, in which the centre sets strategic objectives and steers local governments to conduct policy trials that can later be scaled up nationally, aligning local development agendas with national priorities rather than fully autonomous, bottom-up agenda-setting (Heilmann, 2008). In this context, the innovation outcomes of China’s smart-city pilots are shaped not only by technological adoption, but also by the mobilisation of organisational, fiscal and regulatory resources in line with central mandates, as highlighted in broader analyses of state entrepreneurialism and performance-based incentives in China (F. Wu, 2018).

Theoretical framework. Evaluating the innovation effects of smart-city policies requires not just static benchmarking indicators but an explanatory framework that links policy interventions to the contextual conditions under which they operate. Early industry and policy frameworks—developed, for example, by IBM’s “Smarter Cities” programme, the European Union’s smart-city initiatives, and ISO standards such as ISO 37120—primarily focus on descriptive performance indicators, including broadband coverage, sensor deployment, energy consumption or service accessibility (ISO & IEC, 2019). While these metrics are useful for cross-city comparison, they remain limited in at least three important respects. They remain heavily technology- and output-oriented, privileging infrastructure and service efficiency over organisational capacities, governance processes and local institutional contexts (Angelidou, 2015; Neirotti et al., 2014). They are also largely static and cross-sectional: they record a city’s level of “smartness” at a given moment but say little about the mechanisms through which smart-city strategies generate, or fail to generate, innovation over time (Dolmans et al., 2023; Hollands, 2015). In addition, they tend to treat the city as a homogeneous bundle of assets rather than as a dynamic socio-technical ecosystem in which digital technologies interact with diverse organisations, multi-level governance arrangements and uneven market and regulatory environments (Nastjuk et al., 2022; Rodríguez Bolívar et al., 2023). As a result, indicator-based approaches are ill-equipped to explain why ostensibly similar smart-city projects can yield divergent innovation trajectories across different urban and institutional settings, especially in contexts—such as China—where smart-city programmes operate as state-designed policy pilots rather than purely market-driven initiatives.

To address these analytical limitations, this study adopts the Technology–Organization–Environment (TOE) framework as its primary theoretical lens. The TOE framework conceptualizes innovation adoption and impact as the result of interactions

among three interdependent contexts: the technological base, organizational capabilities, and surrounding environmental conditions (Drazin, 1991). Contemporary empirical scholarship has demonstrated that the TOE framework is particularly efficacious for interrogating complex digital transformations, as it eschews technological determinism in favor of a configurational perspective that explicitly links technology deployment to broader organizational and institutional conditions (Benchis et al., 2025). Within the smart city domain, this perspective is instrumental in framing governance not merely as a technical upgrade, but as a mechanism of institutional change wherein digital infrastructures reconfigure administrative routines and stakeholder relationships to enhance urban livability (Chen et al., 2025). This analytical approach is particularly salient for evaluating China’s national smart city pilot programme, where pilots function as centrally designed policy bundles that reshape local incentive structures; applying the TOE framework enables a granular examination of how unified national policies interact with city-specific technological endowments and environmental pressures to engender differentiated construction efficiency and innovation outcomes (Liao & Chen, 2025).

Smart-city innovation effects and research gaps. Several studies report that smart-city policies can stimulate urban innovation by upgrading ICT infrastructure, enabling digital governance, and enhancing overall city efficiency (Barba-Sánchez et al., 2019a), Barba-Sánchez et al., (2019b); Caragliu & Del Bo, 2019a). Robust digital infrastructure and open-data platforms associated with smart-city projects facilitate rapid information exchange and reduce barriers to innovation, fostering knowledge sharing, entrepreneurial activity, and collaborative creativity (Barba-Sánchez et al., 2019a), Barba-Sánchez et al., (2019b); Du & Wang, 2024). However, other research suggests that performance-assessment regimes can lock smart-city programmes into KPI-driven, short-term outputs, while their policy focus often prioritises immediate service improvements over more fundamental technological breakthroughs (Dolmans et al., 2023). Thus, although enhanced ICT infrastructure and governance processes may raise the sheer quantity of innovative activity, the depth and long-term impact of that activity are less clear, and smart-city projects may achieve only limited progress in terms of technological-innovation outcomes (Caragliu & Del Bo, 2019a).

Regional heterogeneity in policy outcomes further merits attention. Prior work shows that cities differ markedly in innovative capacity owing to variations in economic development, infrastructure maturity, human capital, and governance quality (Rodríguez-Pose et al., 2021; Y. Yang et al., 2023). Highly developed urban areas—benefiting from greater absorptive capacity and infrastructural readiness—may realise more immediate gains from smart-city investments, whereas less-developed regions could experience delayed or divergent effects; some studies even suggest that lagging areas may reap relatively larger benefits given their lower initial baselines (Sun et al., 2023a), Sun et al., (2023b)). Systematic empirical analyses that explicitly examine such regional disparities under smart-city policies, however, remain scarce. Moreover, because smart cities often function as regional hubs, it is still unclear whether pilot-oriented smart-city policies generate positive knowledge and innovation spillovers to neighbouring cities or instead produce a “siphon effect” that concentrates resources and capabilities in pilot areas (M. Wang & Zhou, 2025; T. Wu et al., 2025). Clarifying these spatial dynamics is essential for evaluating the broader regional impact and overall effectiveness of smart-city investments.

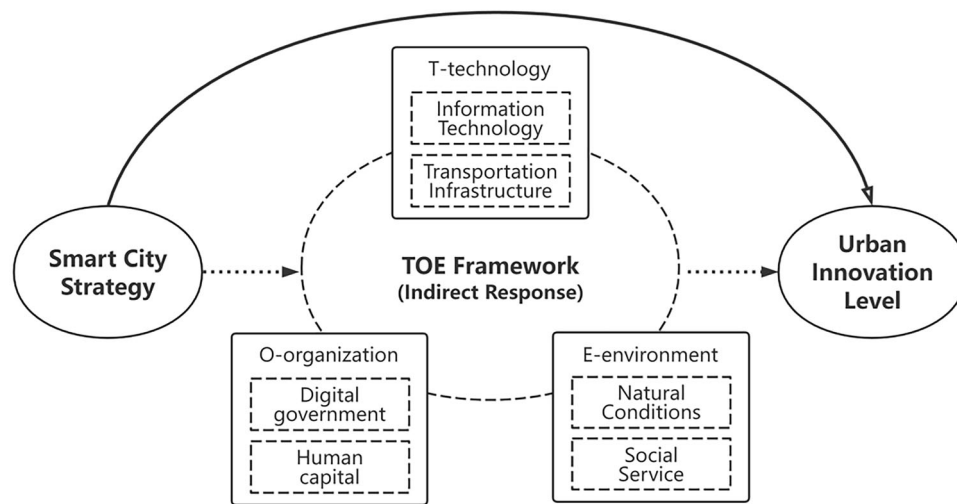


Fig. 1 Theoretical mechanism of smart city strategies impact on urban innovation.

This study seeks to bridge these gaps by systematically evaluating the relationship between China's smart-city pilot policy and urban-innovation-quality. By empirically exploring heterogeneous responses across China's diverse regions and city sizes, as well as potential spatial spillover effects, we offer a comprehensive account of the conditions under which smart-city strategies can effectively foster substantive urban innovation. These insights advance the theoretical understanding of smart-city policy effectiveness and provide valuable guidance for optimising future urban-innovation strategies.

Hypotheses, models and variables

Theoretical framework and hypotheses. As illustrated in Fig. 1, the theoretical model examines both the direct impact of smart city construction on urban innovation levels and the mechanisms through which it operates. The model, which is based on the TOE framework, details how smart city strategies influence urban innovation through the technological, organizational, and environmental dimensions. The technological dimension includes information technology and transportation infrastructure, the organizational dimension encompasses digital government and transportation infrastructure, and the environmental dimension covers the natural and social environments. These elements, when enriched through a smart city strategy, can enhance urban innovation capabilities and determine the policy's effectiveness. This study also considers the spatial spillover effects of smart city construction, addressing the inter-regional innovation impacts arising from policy implementation.

Based on this theoretical model, four research hypotheses are proposed.

Smart cities foster urban innovation and sustainable development by leveraging ICT to provide technical support and facilitate enhanced intelligent city management and services (Agboola et al., 2023; Neirotti et al., 2014). This development accelerates knowledge-sharing and the application of innovative technologies through open data platforms and innovation incubators (Komninos, 2019). The abundant resources, technical support, and conducive environment created by smart city initiatives attract innovative enterprises and high-tech talent, thereby establishing a robust innovation ecosystem (Caragliu et al., 2011; Zeng et al., 2023). Furthermore, smart city development not only upgrades infrastructure and public services but also optimizes resource allocation, stimulates innovation, and promotes entrepreneurship, creating favorable conditions for

innovative and entrepreneurial activities (Maulan et al., 2023; Mitra et al., 2022). Based on these observations, this study proposes the following hypothesis:

Hypothesis 1: Smart city construction positively influences urban innovation levels.

China's vast geography results in significant regional variations in economic development, technological infrastructure, population density, and socio-cultural aspects. This regional heterogeneity leads to differential impacts of smart city construction on urban innovation. For instance, eastern coastal cities, which are characterized by higher economic development and greater openness, are more likely to attract talent and technology, potentially giving them an advantage in promoting urban innovation compared to less developed regions. Additionally, larger cities with greater talent pools and resources, as well as higher population densities, face higher demands and therefore have stronger incentives to innovate, further driving urban innovation (Balland et al., 2020; Shukai et al., 2021). Based on this analysis, the following hypothesis is proposed:

Hypothesis 2: The impact of smart city construction on urban innovation varies across different regions.

The multidimensional characteristics of smart city construction imply a complex and multifaceted impact on urban innovation. From a technological perspective, smart city construction accelerates the adoption of next-generation information technologies such as big data and 5 G, enhancing urban informatization, providing enterprises with abundant innovation resources and technological platforms, and fostering data-driven decision-making and innovation (Bibri & Krogstie, 2020; Gohar & Nencioni, 2021). In terms of organizational collaboration, smart city initiatives optimize human capital structures, attract and concentrate highly skilled talent, and enhance collaboration and communication between firms and government, thus providing robust organizational support for innovation projects (Lim et al., 2023; Pereira et al., 2017). Additionally, smart city initiatives improve urban ecological environments, increase livability, and create favorable conditions for civic citizen well-being and innovation activities (Addas, 2023). Based on these insights, the following hypothesis is proposed:

Hypothesis 3: Smart city initiatives influence urban innovation through technological, organizational, and environmental mediators.

The successful implementation of smart city pilots creates regional centers of knowledge and technology. These pilots often spearhead infrastructure development and high-tech industry growth in the region, endowing neighboring cities with enhanced economic development conditions and innovative environments (Zeng et al., 2023). Moreover, a smart city's construction and policy implementation efforts foster talent concentration and increase innovation activities, which subsequently diffuse to neighboring regions through various networks and partnerships. For instance, cross-city project collaborations enable technological resources and policies to spill over administrative boundaries, promoting co-development among multiple cities in the region through strategic alliances or innovation networks, thereby enhancing the region's overall innovation capacity (Rădulescu et al., 2020). Based on this analysis, the following hypothesis is proposed:

Hypothesis 4: Smart city pilot policies have spatial spillover effects on urban innovation.

Models

Baseline model. The multi-period difference-in-differences (DID) method is a robust econometric tool for assessing the impact of policy changes or events on specific outcomes. We adopt a staggered multi-period DID as the baseline identification strategy, because smart city pilots were introduced in several batches and the design allows us to trace within-city changes in innovation and compare them with those in non-pilot cities over time. Compared with dynamic panel approaches, this framework does not require strong assumptions about the exact lag structure of innovation responses and focuses instead on differences in pre- and post-treatment trends between treated and untreated cities. To mitigate concerns about heterogeneous treatment timing, we use an event-study specification to examine pre-trends and dynamic effects. Spatial dependence is then examined in a subsequent step using a spatial Durbin model. This modelling strategy is consistent with recent quasi-natural experiment studies that evaluate China's digital economy and smart city policies using staggered DID designs (Q. Guo & Zhong, 2022; Sun et al., 2023a), Sun et al., (2023b)). Following the methodology of Callaway and Sant'Anna (2021), this study employs interaction terms to identify policy effects, constructing the multi-period DID model as follows:

$$\ln inn_{it} = \alpha_0 + \alpha_1 DID_{it} + \sum \gamma_j CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (1)$$

Where $\ln inn_{it}$ is the logarithmic form of the explanatory variable of urban innovation level. The core explanatory variable DID_{it} is a dummy variable indicating whether city i implemented a smart city pilot policy in year t . CV_{jit} represents control variables affecting the city's innovation level in year t . Crucially, to address endogeneity concerns stemming from omitted variable bias, the model incorporates high-dimensional fixed effects: μ_{city} absorbs time-invariant unobserved heterogeneity (e.g., geographical location and cultural endowment), v_t captures common macro-economic shocks, and λ_{it} is the random error term.

Mediation effects model. This study constructs a mediation model to investigate the mechanisms through which smart city pilots influence urban innovation levels. Initially, a series of mediating variables are treated as dependent variables, with smart city pilot as the primary explanatory variable in a regression analysis to test its impact on these mediating factors. If the estimated coefficient of smart city pilot is significant, it indicates that the smart city pilot variable affects the mediating variables. Next, both the mediating variables and the smart city pilot variable are included

in the model as explanatory variables, with the urban innovation index as the dependent variable. If the coefficients of both the pilot variable and the mediating variables are significant, this suggests that the mediating variables play a partial mediating role in the effect of a smart city pilot program on urban innovation. If the coefficient of the pilot variable is not significant but the coefficient of the mediating variable is, this indicates that the mediating variable has a full mediating effect. The mediation effect model is specified as follows:

$$M_{it} = \beta_0 + \beta_1 DID_{it} + \sum \delta CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (2)$$

$$\ln inn_{it} = \beta_2 + \beta_3 DID_{it} + \beta_4 M_{it} + \sum \gamma_j CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (3)$$

M represents the mediating variable, and the other variables are defined as previously mentioned.

Spatial econometric model. Cities do not exist in isolation, and they interact with and depend on each other economically, socially, and technologically. Ignoring spatial interactions between regions when examining the effects of smart city pilot policies can thus lead to inaccurate and unreliable conclusions. Therefore, this paper tests for spatial spillover effects to determine whether the impacts of smart city pilots on urban innovation extend to neighboring regions through various channels. First, the Lagrange multiplier test, Hausman test, and likelihood ratio test are conducted. Based on the results and following the study by Wang L. and Shao (2024), a difference-in-differences spatial Durbin model is introduced as follows:

$$W_{ij} = \begin{cases} \frac{1}{d_{ij}^2}, i \neq j \\ 0, i = j \end{cases} \quad (4)$$

$$Y_{it} = \theta_0 + \theta_1 DID_{it} + \theta_2 W * INN_{it} + \theta_3 W_{ij} DID_{it} + \sum_{j=1}^n \gamma_j W_{ij} CV_{it} + \mu_{city} + v_t + \lambda_{it} \quad (5)$$

Where W is the spatial weight matrix. Due to data collection challenges and the exclusion of non-compliant cities, the selected sample cities are not completely contiguous. Therefore, an inverse geographic distance spatial matrix is constructed using the inverse square of the geographic distance between cities, as shown in equation (4). $W * INN_{it}$, $W_{ij} DID_{it}$ and $W_{ij} CV_{it}$ denote the spatial lag terms of the dependent, independent, and control variables, respectively. Y_{it} represents the dependent variable; θ_0 is the constant term; θ_1 represents the regression coefficients of the independent variables; θ_2 , θ_3 and γ_j denote the spatial lagged regression coefficients of the dependent, independent, and control variables, respectively. The remaining symbols retain their meanings as defined in equation (1).

Variables

Dependent variable. Considering that previous studies have typically relied on indicators such as financial investment in science and technology or the number of patents granted, which may not fully capture the breadth of urban innovation, this study adopts a more comprehensive measure, the Urban Innovation Index, to assess a city's innovation capacity. Traditional metrics such as patent counts are informative, but they generally ignore heterogeneity in patent quality, and financial indicators capture innovation inputs rather than realized outputs. Accordingly, we use the Urban Innovation Index developed by the Fudan Institute of Industrial Development, which is constructed from invention patent data provided by the China National Intellectual Property Administration and a structural patent-renewal model that

Table 1 Indicators of the intermediary variables.			
TOE Framework	Primary Indicator	Secondary Indicator	Original Indicator Data
Technical Conditions	Information and Communication Technology	Internet Penetration	Number of International Internet Users per 100 People
		Communication Penetration	Mobile Phone Penetration Rate
		Information Infrastructure	Long-Distance Optical Cable Density
	Transportation Infrastructure	Internet Infrastructure	Per Capita Broadband Internet Access Ports
		Road Construction	Per Capita Road Area
		Road Coverage and Connectivity	Highway Mileage
		Rail Transit	Number of High-Speed Rail Stations Opened
Organizational Conditions	Digital Government	Public Transport Benefit	Number of Public Transport Vehicles per 10,000 People
		Digital Intelligence Emphasis	Frequency of Government Work Report
		Government Transparency	Government Fiscal Transparency
		Scientific Support	Government Science Expenditure
	Human Capital	Government Performance	Government Website Performance Evaluation Index
		High-level Talent Efficacy	Per capita gross output value of employees in high-tech zones
		Citizen Income Level	Per Capita Disposable Income
Environmental Conditions	Urban Natural Environment	Education Level	Number of Students in Higher Education per 1,000 People
		Scientific and Technological Talent Resources	Proportion of Employees in Information Transmission, Computer, and Software Industries
		Urban Air Quality	Annual Average PM2.5 Concentration
		Urban Environmental Attractiveness	Per Capita Park Green Area
	Social Service Environment	Urban Water Source Status	Centralized Treatment Rate of Sewage Treatment Plants
		Environmental Governance	Environmental Regulation Strength
		Basic Education	Student-teacher ratio in regular secondary schools
	Medical and Health Care	Number of hospital beds per 100 people	
	Cultural Services	Number of public library books per 10,000 people	
	Social Security	Proportion of participants in endowment insurance to the year-end total population	

assigns age-specific value weights to granted patents (Kou & Liu, 2017). In each year, the estimated values of all valid invention patents are aggregated at the city level to obtain the Urban Innovation Index, which provides a more comprehensive and quality-adjusted measure of urban innovation.

Independent variables. The primary explanatory variable in this study is smart city pilot policy, defined as a dummy variable. For cities designated as pilots, the variable is set to 1 for the year the policy is implemented and subsequent years; for non-pilot cities or years before policy implementation, it is set to 0.

Control variables. To mitigate omitted variable bias, we adopt the Regional Innovative Capacity framework (Furman et al., 2002), which attributes regional innovation performance to three dimensions: common innovation infrastructure, the cluster-specific environment, and the quality of linkages. First, regarding the common innovation infrastructure, we control for Economic Development Level and City Classification. Consistent with the RIC framework, a strong economic foundation (measured by log per capita GDP) provides the resources necessary to buffer R&D risks (Furman et al., 2002). In China’s hierarchical administrative system, high-ranking cities typically enjoy expanded powers relative to ordinary prefecture-level cities; we therefore include a dummy variable to capture these institutional advantages (Lu & Tsai, 2021). Second, to capture the cluster-specific environment, we control for Urban Agglomeration, measured by log population density. Consistent with agglomeration theory, higher spatial density reduces

knowledge transmission costs and fosters the cross-fertilization of ideas by facilitating face-to-face interactions and labor matching (Matsiuk et al., 2024). Third, reflecting the quality of linkages to the global frontier, we control for the Level of Openness, measured by the ratio of FDI to GDP. FDI acts as a conduit for international technology transfer, promoting domestic innovation through demonstration effects and labor mobility (Y. Wang et al., 2025).

Mediator variables. This study utilizes the TOE theoretical framework, as depicted in Fig. 1, encompassing technology, organization, and environment components. This framework is designed to explore the interactions and causal relationships among technological innovations within a specific context. Technological advancements can foster new forms of urban innovation, organizational capabilities determine the efficiency and direction of innovation activities, and environmental factors shape the opportunities and constraints for urban innovation. Hence, this theoretical framework is well-suited to the study’s theme. Drawing on established literature, we align the TOE components with the six-dimension smart-city construct (Table 1): technological conditions (transport and ICT infrastructure), organizational conditions (human capital and digital government), and environmental conditions (social and natural resources)(Caragliu et al., 2011; Marchesani & Ceci, 2025; Shao & Min, 2025).

Data and descriptive statistics. Considering the timeline of policy implementation and data availability, this study selects 283

Table 2 Descriptive statistics for key variables.							
Variable	Variable name	Variable symbol	N	mean	sd	min	max
Explained Variables	Urban Innovation Index	CII	4528	0.650	2.046	−5.273	7.975
Explanatory Variables	Smart City Pilot	DID	4528	0.207	0.405	0	1
Control Variables	City Clustering	PD	4521	5.727	0.941	0.446	7.882
	City Ranking	CC	4528	0.106	0.308	0	1
	Economic Development	ED	4516	10.49	0.724	4.595	13.06
	Openness to the Outside World	FO	4490	1.706	1.498	0.161	19.783

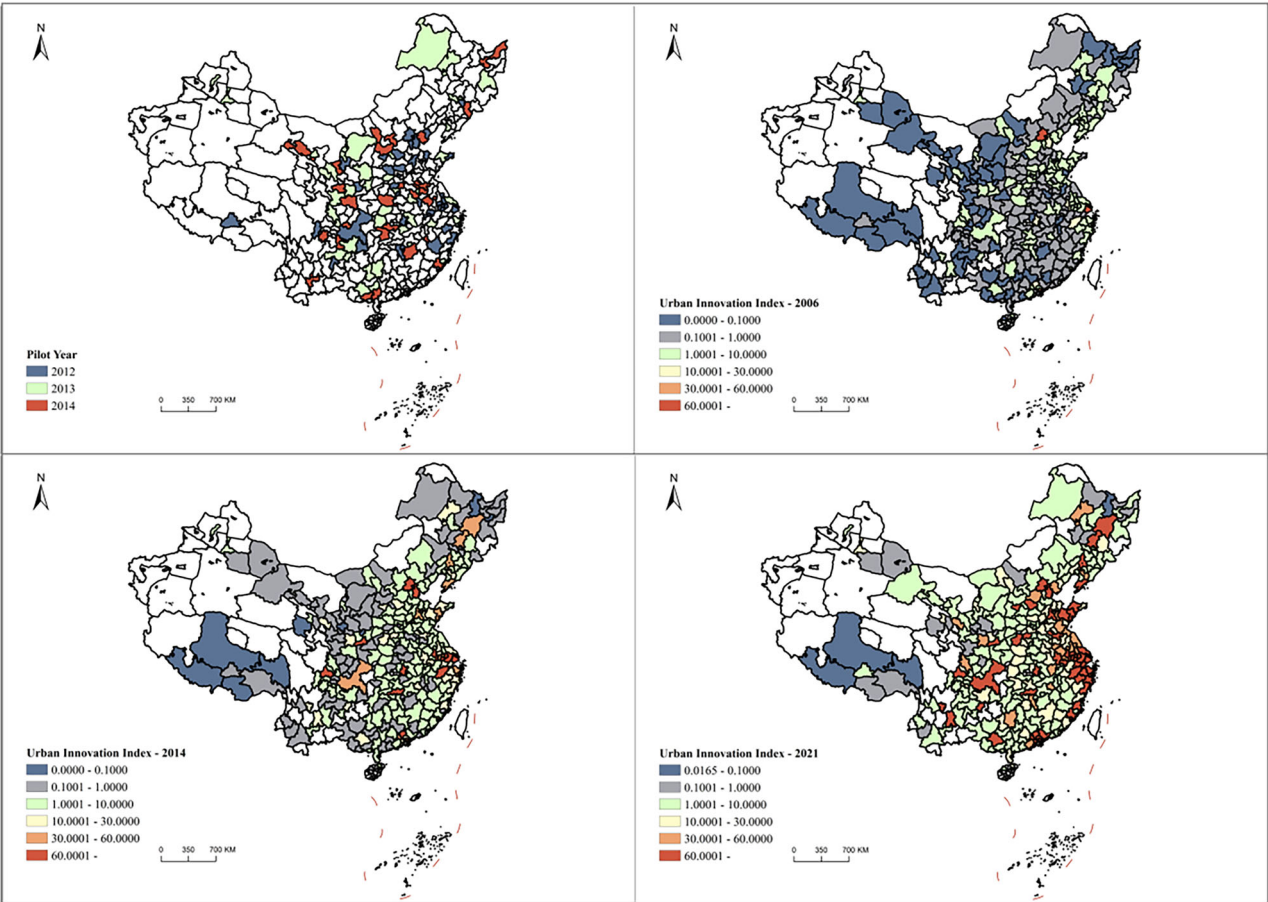


Fig. 2 China’s smart city pilots and urban innovation index in 2006, 2014, 2021.

prefecture-level cities in China from 2006 to 2021 as the research sample. The variables used in the model and their descriptive statistics are presented in Table 2. Data sources include the “Report on the Innovative Power of Chinese Cities and Industries” by Fudan University, the three batches of the “Smart City Pilot List” from the Ministry of Housing and Urban-Rural Development, the China Land Market Network, the “Statistical Yearbook of Land and Resources,” and Peking University’s Digital Financial Inclusion Index. Data for the control and mediating variables are sourced from various authoritative statistical yearbooks and databases.

As shown in Fig. 2, smart city pilots are primarily concentrated in the central and eastern regions, especially in eastern China. From 2006 to 2021, the urban innovation index has shown an overall upward trend, with the eastern region exhibiting higher innovation capacity compared to the central, western, and northeastern regions. This “high in the east, lower in the west” distribution pattern of innovation levels is similar to the distribution of smart city pilots, indicating that smart city

projects may be having a positive effect on enhancing urban innovation capacity.

Empirical results

Baseline regression. Prior to the baseline estimation, we verified the parallel trend assumption to ensure the validity of the multi-period DID strategy (Figure A1). The event study results indicate that the treatment and control groups exhibited similar pre-treatment trends. Subsequently, the impact of smart city pilot construction on urban innovation levels is analyzed using equation (1), with the benchmark regression results shown in Table 3. The first column presents the impact of smart city pilot policies on urban innovation without the control variables, while column (2) shows the results with the control variables. Both models account for city and time fixed effects. The results indicate that the estimated coefficient for the explanatory variable is 0.156 without the control variables and 0.121 with the control variables, both significant at the 1% level. These results suggest that smart

city pilot construction is positively associated with urban innovation capacity, supporting Hypothesis 1. The models' R^2 values of 0.972 and 0.973 indicate a high degree of goodness of fit. This may be attributed to the fact that intelligent manufacturing and smart governance—pivotal components of smart cities—provide policy support for intelligent transformation and industrial upgrading, reduce information asymmetry, and optimize resource allocation, thereby fostering collaborative innovation in cities (Branco et al., 2023; Jin & Chen, 2024).

In addition, placebo tests, alternative sample periods and innovation measures, outlier trimming, and controls for other major national urban policies (reported in Appendix Table A2-4) yield similar results, supporting the robustness of these baseline estimates.

The analysis of control variables shows that higher population density, capital city status and stronger economic development

are associated with higher urban innovation. However, foreign investment – a proxy for openness – negatively impacts urban innovation. This might be because foreign direct investment is often concentrated in highly profitable sectors rather than innovative ones, leading to uneven regional innovation stimulation. Investments prioritizing efficiency and short-term gains over long-term innovation development exacerbate this issue. Moreover, foreign investment can lead firms to rely excessively on external support, with high-tech companies potentially prioritizing the acquisition of advanced foreign technologies over independent R&D (Li, 2011). Additionally, when foreign capital repatriates more profits to its home country and reduces investment in the host country, it can diminish the host country's R&D and innovation capacity (Wang Y. et al., 2007).

Heterogeneity analysis. China is a vast country with prominent issues of unbalanced and insufficient regional development, resulting in significant disparities between cities (Hu & Zheng, 2021; Zhuang & Ye, 2020). Therefore, this paper conducts a heterogeneity test along two dimensions: location and size, as detailed in Table 4. First, there are notable differences between eastern China and central and western China regarding economic development levels, infrastructure development, policy support strength, and openness (Yu & Shang, 2018). Based on the geographical division standard published by the National Bureau of Statistics, the sample cities are divided into eastern and central-western regions. As shown result, smart city construction is statistically significantly associated with higher urban innovation in the central and western regions, but no statistically significant effect was observed in the eastern region. This suggests that in the eastern region, where innovation ecosystems are already mature, the marginal impact of the pilot policy may be limited (Hu & Zheng, 2021; Zhuang & Ye, 2020). In contrast, the less developed central and western regions appear to benefit more from smart city projects, as new technologies and ideas introduced during construction boost urban innovation.

Second, considering the heterogeneity of city sizes (Zhang et al., 2022), the sample cities are categorised into large cities and small to medium-sized cities according to the “Notice on

Table 3 Benchmark regression results.

VARIABLES	(1) CII	(2) CII
DID	0.156*** (0.022)	0.121*** (0.022)
PD		0.110*** (0.031)
CC		0.064** (0.026)
ED		0.254*** (0.030)
FO		−0.018*** (0.005)
Constant	0.625*** (0.007)	−2.635*** (0.353)
Observations	4528	4512
R-squared	0.972	0.973
Time fixed effects	YES	YES
Individual fixed effect	YES	YES

Note: **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Table 4 Heterogeneity test results.

VARIABLES	City Location		City Size		
	Midwest Region CII	Eastern Region CII	Small and Medium-sized Cities CII	Large Cities CII	Add to Interaction CII
DID	0.160*** (0.036)	0.063 (0.042)	0.189*** (0.039)	0.096* (0.051)	0.076** (0.036)
MCDID					0.111*** (0.036)
PD	0.078** (0.033)	−0.197 (0.161)	0.042 (0.051)	0.140*** (0.036)	0.108*** (0.031)
CC	0.094* (0.049)	0.066* (0.035)	0.089** (0.038)	0.057* (0.031)	0.062** (0.026)
ED	0.328*** (0.035)	0.221*** (0.055)	0.308*** (0.039)	0.225*** (0.049)	0.257*** (0.030)
FO	0.026*** (0.006)	−0.044*** (0.007)	0.010 (0.008)	−0.034*** (0.006)	−0.017*** (0.005)
Constant	−3.709*** (0.390)	0.650 (1.248)	−3.612*** (0.456)	−1.102* (0.577)	−2.658*** (0.353)
Observations	3117	1375	2896	1596	4512
R-squared	0.965	0.983	0.951	0.975	0.973
Time fixed effects	YES	YES	YES	YES	YES
Individual fixed effect	YES	YES	YES	YES	YES

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Adjusting the Criteria for the Classification of City Scale.” The regression analysis suggests that smart city construction is more strongly associated with innovation in small and medium-sized cities than in large cities. Specifically, the coefficient of the interaction term between population size and the pilot variable is significantly positive. This phenomenon may be due to the Chinese government’s recent emphasis on coordinated urban economic development, promoting the growth of small to medium-sized cities through policy and financial support, allowing these cities to benefit more from smart city construction (Yu & Shang, 2018; Liu et al., 2022). Additionally, smart city construction in small to medium-sized cities can effectively attract talent, promote knowledge exchange and technology transfer, and accelerate the development and improvement of the urban innovation system. Conversely, in large cities, the high population density results in a smaller marginal effect of smart city construction on talent attraction.

In summary, the heterogeneity test reveals differentiated associations between smart city construction and innovation across regions and city sizes, with relatively stronger effects in central and western regions and in small and medium-sized cities. Beyond these two dimensions, we further examine heterogeneity by administrative hierarchy, fiscal capacity and industrial structure; the corresponding estimates are reported in Appendix Table A1.

Mediation test. Based on the theoretical framework and the validated constructs above, this section analyzes the internal mechanisms by which smart city pilot construction influences urban innovation capacity. Following the methodology of Cargliu and Del Bo (2019), the four secondary indicators in each category in Table 1 are subjected to principal component analysis to extract composite indicators. Each composite indicator is then treated as a mediating variable for the mediation effect test. The results of the heterogeneity analysis reveal that the impact of smart city pilots on urban innovation varies by geographic location, but the differences due to city size are less pronounced. Therefore, this mechanism analysis focuses on the geographic location differences. The mediation mechanism analysis is based on the TOE framework outlined in Section “Theoretical framework and hypotheses”.

Crucially, to address potential concerns that mediating variables within the TOE framework (e.g., ICT infrastructure and digital government) might represent redundant underlying processes, we conducted a collinearity diagnostic prior to the regression analysis. We generated a correlation matrix heatmap (Appendix Fig. A2) to visually inspect the relationships between mediators. Furthermore, the Variance Inflation Factor (VIF) results (see Appendix Table A5) show a mean VIF of 2.18 and a maximum of 4.06, well below the critical threshold of 10. These statistics, combined with the theoretical distinction between technical connectivity (ICT, transportation) and organizational capabilities (digital government, human capital), confirm that the mediators capture distinct mechanisms rather than alternative measures of the same construct.

The results of the heterogeneity analysis reveal that the impact of smart city pilots on urban innovation varies by geographic location, but the differences due to city size are less pronounced. Therefore, this mechanism analysis focuses on the geographic location differences. The mediation mechanism analysis is based on the TOE framework outlined in Section “Theoretical framework and hypotheses”.

Technological effect. The technological effect is divided into two aspects: ICT and transportation infrastructure. The results in

Table 5 Results of the mediation test for technological conditions.

VARIABLES	Information Technology			Eastern Information Technology			Central and Western Information Technology			Transportation Infrastructure			Eastern Transportation Infrastructure			Central and Western Transportation Infrastructure		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(9)	(10)	(11)	(12)	(11)	(12)
	IT	CII	IT	CII	IT	CII	TI	CII	TI	CII	TI	CII	TI	CII	TI	CII	TI	CII
DID	0.292*** (0.111)	0.120*** (0.022)	0.104 (0.258)	0.064* (0.034)	0.308** (0.128)	0.161*** (0.026)	0.937*** (0.213)	0.108*** (0.022)	0.224 (0.363)	0.063* (0.033)	1.150*** (0.259)	0.144*** (0.026)	control	control	control	control	control	control
Mediating variables	control	control	control	control	control	control	control	control	control	control	control	control	control	control	control	control	control	control
Constant	15.838*** (1.930)	-2.681*** (0.356)	82.939*** (9.646)	0.110 (1.282)	13.575*** (1.926)	-3.685*** (0.393)	7.973** (3.501)	-2.640*** (0.354)	6.218*** (2.426)	-0.040 (1.247)	8.442** (3.911)	-3.738*** (0.391)	control	control	control	control	control	control
Observations	4506	4492	1388	1375	3118	3117	4475	4474	1373	1373	3102	3101	control	control	control	control	control	control
R-squared	0.884	0.973	0.784	0.983	0.897	0.965	0.892	0.973	0.825	0.983	0.906	0.966	control	control	control	control	control	control
Time fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Individual fixed effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Table 6 Results of the mediation test for organizational conditions.

VARIABLES	Digital Government		Eastern Digital Government		Central and Western Digital Government		Human Capital		Eastern Human Capital		Central and Western Human Capital	
	(1) DG	(2) CII	(3) DG	(4) CII	(5) DG	(6) CII	(7) HC	(8) CII	(9) HC	(10) CII	(11) HC	(12) CII
DID	0.413*** (0.078)	0.117*** (0.022)	0.038 (0.139)	0.064* (0.034)	0.562*** (0.095)	0.158*** (0.026)	0.069* (0.040)	0.114** (0.052)	0.254*** (0.082)	0.045** (0.021)	0.026 (0.031)	0.162** (0.078)
Mediating variables		0.009** (0.004)		0.003 (0.007)		0.004** (0.002)		0.105** (0.042)		0.078** (0.036)		0.005* (0.003)
Control variable	control	control	control	control	control	control	control	control	control	control	control	control
Constant	1.421 (1.285)	−2.648*** (0.353)	1.055 (5.208)	0.645 (1.248)	0.116 (1.426)	−3.710*** (0.390)	4.573*** (0.563)	−3.210*** (1.086)	3.962** (1.850)	0.335* (0.195)	3.836*** (0.515)	−3.818*** (1.226)
Observations	4506	4492	1388	1375	3118	3117	4506	4492	1388	1375	3118	3117
R-squared	0.873	0.973	0.720	0.983	0.876	0.965	0.924	0.971	0.926	0.982	0.923	0.962
Time fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Individual fixed effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Table 5 indicate that, overall, smart city construction significantly promotes the development of ICT and transportation infrastructure at the 1% level. The enhancement of these technological aspects, in turn, is associated with higher urban innovation levels. Sub-regional regression results show that in the eastern region, technological conditions do not mediate the effect of smart cities on urban innovation. However, in the central and western regions, smart city construction can enhance urban innovation by advancing technological development.

The differences between the aforementioned regions may be attributed to the more mature technological development in the eastern region, leading to smaller marginal benefits from a smart city pilot policy. Conversely, the central and western regions, with relatively lagging technological infrastructure, can benefit significantly from the high-tech advancements and management models introduced by a smart city pilot policy. Furthermore, smart city construction significantly improves transportation accessibility in the central and western regions through the implementation and promotion of smart transportation. A robust transportation system not only enhances urban connectivity but also facilitates the efficient flow of information, talent, and resources.

Organizational effect. The mediation results for organisational conditions reported in Table 6 indicate that digital government and human capital together form an important channel linking the smart city strategy to urban innovation. In the full sample, improvements in digital government significantly transmit policy effects to higher innovation, while the human capital mediator is also positive but becomes weaker and sometimes loses significance once additional controls are introduced. In the eastern region, the estimated effect of human capital is larger and more robust than that of digital government, whereas the organisational channel is clearly weaker in central and western cities. This pattern suggests that eastern cities, which host dense clusters of high-tech firms and well-developed labour markets, rely more strongly on market-based mechanisms to translate smart city investments into better matching between skilled workers and innovative enterprises. Smart city projects that enhance data availability, transparency and online services reinforce these market processes and increase the attractiveness of leading cities

for mobile talent and innovative companies (Woods et al., 2024), while at the same time raising questions about the social consequences of entrepreneurial, technology-led urban development (Khurana & Dutta, 2024).

By contrast, in central and western cities the mediating effect of digital government is statistically significant, while the human capital channel is small and often not significant. This suggests a threshold effect: in regions where high-tech industries and skilled labour pools exist but are less concentrated and innovation networks are thinner, smart city initiatives first tend to strengthen basic administrative capacity, information transparency and online public services. The productivity gains associated with more efficient allocation of human resources are likely to materialise more gradually as industrial clusters deepen and local innovation systems mature. In these contexts, the main organisational contribution of the smart city strategy is to improve government coordination and reduce transaction costs, which is consistent with evidence that more capable smart city governance and open, multi-actor innovation ecosystems create more favourable conditions for inclusive and innovation-oriented urban development (Kummitha, 2018b; Mora et al., 2025).

Environmental effect. Regarding the environmental effects, this study focuses on natural conditions and social services, with the test results shown in Table 7. The mediating role of environmental conditions is comparatively modest in the baseline results, and the overall coefficient is not robustly significant. This outcome can be understood in light of the fiscal and temporal characteristics of investments in environmental quality and public service provision: smart city projects often prioritise digital platforms and visible information infrastructure, whereas improvements in environmental indicators and basic services require long-term budget commitments and cross-sector coordination. Recent work on smart cities and inequality finds that such initiatives can alleviate disparities, but that their distributional effects are heterogeneous and depend on whether policies explicitly address social equity alongside technological upgrading (Caragliu & Del Bo, 2022; Z. Yang et al., 2024).

At the same time, the regional estimates indicate that environmental conditions play a more pronounced mediating role in central and western cities, where the coefficient for the

Table 7 Results of the mediation test for environmental conditions.

VARIABLES	Natural Conditions		Eastern Natural Conditions		Central and Western Natural Conditions		Social Services		Eastern Social Services		Central and Western Social Services	
	(1) NC	(2) CII	(3) NC	(4) CII	(5) NC	(6) CII	(7) SS	(8) CII	(9) SS	(10) CII	(11) SS	(12) CII
DID	0.690** (0.340)	0.119*** (0.022)	2.971*** (0.779)	0.074** (0.034)	0.017 (0.356)	0.160*** (0.026)	0.047 (0.040)	0.106*** (0.035)	−0.037 (0.079)	0.077** (0.036)	0.079* (0.046)	0.138*** (0.046)
Mediating variables		0.003*** (0.001)		0.003** (0.001)		0.003*** (0.001)		0.015** (0.007)		0.041* (0.023)		−0.034* (0.020)
Control variable	control	control	control	control	control	control	control	control	control	control	control	control
Constant	14.265** (5.583)	−2.677*** (0.353)	70.108** (29.103)	0.876 (1.248)	−2.218 (5.356)	−3.701*** (0.389)	6.840*** (0.848)	−2.854** (1.176)	18.079*** (4.218)	0.512* (0.299)	6.137*** (0.894)	−3.091** (1.306)
Observations	4506	4492	1388	1375	3118	3117	4485	4484	1371	1371	3114	3113
R-squared	0.880	0.973	0.933	0.983	0.642	0.965	0.903	0.979	0.946	0.986	0.824	0.968
Time fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Individual fixed effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are Standard errors.

Table 8 Moran's Index.

Research Item	Moran's I	z-value	p-value	Research Item	Moran's I	z-value	p-value
Y2006	0.045	4.8	0.000	Y2014	0.076	6.4	0.000
Y2007	0.047	5.2	0.000	Y2015	0.08	6.2	0.000
Y2008	0.051	5.6	0.000	Y2016	0.086	6.6	0.000
Y2009	0.061	6.0	0.000	Y2017	0.086	6.8	0.000
Y2010	0.063	6.4	0.000	Y2018	0.09	7.0	0.000
Y2011	0.064	6.2	0.000	Y2019	0.095	7.2	0.000
Y2012	0.068	6.6	0.000	Y2020	0.094	7.0	0.000
Y2013	0.069	6.8	0.000	Y2021	0.099	6.6	0.000

social service environment is positive and statistically significant. This pattern is consistent with a compensatory effect: in regions with a historically weaker provision of public services, the smart city pilot is more likely to be accompanied by efforts to expand basic services and improve the local living environment, which helps to retain and gradually attract skilled workers. Research on urban amenities shows that cities investing in the quality of everyday services are better able to attract and keep high-skilled residents, thereby supporting local innovation capacity (Glaeser et al., 2001). At the same time, the negative coefficient observed in some specifications suggests that in fiscally constrained cities, short-term increases in spending on public services may crowd out resources that could otherwise be devoted to research and development, underlining the importance of balancing investments in digital infrastructure, environmental quality and social services within broader urban innovation strategies.

Taken together, the three mediating mechanisms suggest that China's smart city strategy currently operates in a predominantly technology- and efficiency-oriented mode. Technological conditions appear to act as the most stable and sizeable transmission channel from policy pilots to urban innovation across all regions. Organisational conditions, especially human capital and digital government, contribute more strongly in eastern cities, where market institutions and high-tech ecosystems are already relatively mature, while their mediating role is weaker in central and western regions. Environmental and social service conditions remain the least developed mechanism, with limited and heterogeneous effects that are stronger in less-developed regions and in some cases associated with short-

term resource crowding-out. These results are consistent with comparative evidence that smart city policies generate stronger innovation gains when they are embedded in supportive institutional settings, coupled with long-term investments in human and social capital and framed by robust governance arrangements (Caragliu & Del Bo, 2019b; Kummitha & Crutzen, 2017; Mora et al., 2023b).

Spatial spillover effect test

Spatial auto-correlation. Using spatial econometric models for empirical analysis requires an initial assessment of the spatial correlation of the dependent variable. Moran's I statistic is employed to test the spatial auto-correlation of the innovation indices for the 283 Chinese cities from 2006 to 2021. The results in Table 8 reveal significant positive spatial correlation between cities' innovation levels, with a clear geographic clustering effect, indicating substantial spatial spillover effects. The global Moran's I increases annually from 2006 to 2021, suggesting increased spatial dependence. This may be due to cities becoming more interconnected through improved transportation networks and digital communication infrastructure, facilitating the spillover of innovation activities to neighboring cities. Additionally, regional development strategies have strengthened interregional linkages by creating channels for knowledge and resource sharing (Lopes & Franco, 2019; Qin, 2022).

This study follows J. P. Elhorst's approach for model selection and validation (Elhorst, 2014). The spatial Lagrange multiplier (LM) test results show that the p-values are significant at the 1% level, indicating the presence of spatial error and spatial lag

Table 9 Spatial Panel Durbin Model Spillover Effects and Decomposition.

VARIABLES	(1) Main	(2) Wx	(3) Direct	(4) Indirect	(5) Total
DID	0.353*** (0.025)	0.464* (0.272)	0.438*** (0.038)	0.256*** (0.095)	0.694*** (0.102)
PD	0.290*** (0.027)	0.215 (0.448)	0.322*** (0.036)	0.280*** (0.105)	0.602*** (0.111)
CC	0.021* (0.012)	-0.040 (0.029)	0.025* (0.014)	0.032 (0.025)	0.057** (0.029)
ED	0.305*** (0.019)	0.249* (0.155)	0.354*** (0.023)	0.410*** (0.120)	0.764*** (0.122)
FO	-0.095*** (0.029)	-0.243*** (0.050)	-0.107*** (0.030)	-0.285** (0.117)	-0.392*** (0.121)
rho	0.387*** (0.052)				
sigma2_e	0.185*** (0.004)				
Observations	4528	4528	4528	4528	4528
R-squared	0.572	0.572	0.572	0.572	0.572
Number of CITY	283	283	283	283	283

Note: *, **, and *** indicate statistical significance at the 10%, 5%, and 1% confidence levels, respectively. Values in parentheses are standard errors.

effects, leading to the initial selection of the spatial Durbin model, which incorporates both effects. Subsequent post hoc tests, including the Hausman test, confirm that the spatial regression model applies to fixed effects. Finally, the likelihood ratio (LR) test and Wald test results reject the null hypothesis of spatial Durbin model degradation at the 1% significance level, further validating the model's applicability.

Spatial durbin model estimation results. The spatial spillover effects are tested using the difference-in-differences spatial Durbin model, with the results presented in Table 9. Column 1 indicates that the spatial autoregressive coefficient for urban innovation, ρ , is significantly positive at the 1% level, demonstrating the positive spatial dependence of urban innovation. The results show that the smart city pilot policy is statistically significantly associated with higher innovation capacity in neighbouring cities, exhibiting notable spatial spillover effects and verifying Hypothesis 4. The decomposition of the total effect reveals that both direct and indirect effects are significantly positive, indicating that the implementation of the smart city pilot policy not only promotes innovation within the city but also exerts an influence on surrounding cities. Neighboring cities can learn from the pilot regions' experiences in high-tech applications, urban governance models, and talent development strategies to bolster their own innovation capacity and development potential. This cross-regional dissemination of knowledge and experience fosters overall technological advancement and economic growth, enhancing regional sustainable development and the construction of innovation ecosystems (Bevilacqua et al., 2021; Stöber et al., 2023).

Although smart city strategies have generated positive spillovers, the spatiotemporal evolution depicted in Fig. 2 reveals a persistent innovation divide. Inland regions adjacent to high-innovation coastal clusters and Northeast regions near the capital have not manifested the anticipated trickle-down effect, remaining at low innovation levels. In contrast, certain non-adjacent cities with robust economic foundations exhibit synchronized advancement. This visual evidence aligns with the robustness check based on the economic-geographic nested matrix (see Appendix Table A6), and suggests that spillover effects are conditioned by economic similarity. This implies that while geographical proximity serves as a transmission conduit,

absorptive capacity is the critical determinant of reception efficacy (Miguélez & Moreno, 2015a, Miguélez & Moreno, 2015b; Stettler et al., 2025). Given that similar stages of economic development typically correspond to compatible institutional environments and governance capacities (Rodríguez-Pose, 2013), the low-innovation traps observed in Fig. 2 can be attributed to institutional mismatches; specifically, less developed neighbors struggle to effectively interface with the smart systems of pilot cities due to a lack of compatible digital governance infrastructure or administrative efficiency. Where significant governance gradients exist, the siphon effect may overshadow the spillover effect, highlighting that optimizing regional institutional arrangements to reduce barriers to knowledge flow is as critical as investment in physical infrastructure (Mora et al., 2023a).

Conclusions and limitations

Conclusions and discussion. Drawing on a quasi-natural experiment of smart city pilots, this paper combines theoretical analysis with empirical research, utilizing panel data from 283 prefecture-level cities across China from 2006 to 2021. Empirical studies in Europe and other regions have often evaluated smart city policies through cross-sectional comparisons between cities with and without such initiatives, which provides static evidence on innovation outcomes but offers limited insight into dynamic policy effects (Barba-Sánchez et al., (2019a), Barba-Sánchez et al., (2019b)). By employing a multi-period difference-in-differences design, this study instead traces how urban innovation evolves before and after pilot designation and contrasts these trajectories with those of non-pilot cities. It further combines the Technology-Organization-Environment (TOE) framework with mediation analysis to examine the channels through which smart city pilots influence innovation and embeds this setting in a spatial Durbin model to identify spillover effects on neighbouring cities.

First, smart city pilot policies are positively associated with urban innovation, with notable regional heterogeneity. The baseline regressions show a robust and statistically significant positive impact of pilot designation on innovation, which remains stable under multiple robustness checks. However, the effects of these policies vary by region and city size. In the central and western regions, the estimated coefficients are larger and

statistically significant, echoing broader findings that digital and innovation-oriented policies can especially enhance performance in structurally weaker or less diversified cities (Sun et al., (2023a), Sun et al., (2023b)). In contrast, the estimated effects for eastern cities are smaller and statistically insignificant. This implies that where the local innovation base is strong and governance modes rely more heavily on competitive markets and private initiative, the additional marginal value of the national pilot programme may be limited. Regarding city size, smart city policies improve urban innovation across all types of cities, with particularly pronounced effects in small and medium-sized cities. Additional heterogeneity tests also indicate that cities with more diversified industrial structures respond more strongly to the pilot, which underlines the importance of both governance modes and the existing innovation base in translating smart city strategies into measurable innovation gains.

Second, the results show that smart city strategies operate through technological, organisational and environmental conditions, with their relative importance varying across regions. At the aggregate level, the mediation analysis indicates that improvements in technological conditions, organisational capacities and natural environmental indicators are all associated with higher innovation, whereas the mediating role of social service is weak. For eastern cities, the most important mediating channels are human capital and natural environmental quality, while the technological mediator is relatively modest, which differs from many empirical studies that emphasise the role of digital infrastructure and ICT adoption in driving innovation (Du et al., 2023). A plausible interpretation is that, in technologically advanced eastern cities where digital infrastructure is already dense, incremental infrastructure upgrades contribute less to innovation than policies that reshape skills, environmental quality and governance modes—an evolution that aligns with arguments that mature smart cities increasingly adopt human-centred and governance-oriented approaches to development (T. Ji et al., 2021).

In central and western regions, by contrast, technological upgrading and digital government constitute the main mediating channels, while the indirect effect of human capital is not significant and the coefficient on the social-service environment is negative. This pattern suggests that, at the current stage, smart city investments in these areas are still concentrated on basic digital infrastructure, administrative modernisation and core public services, and in some cases may temporarily divert fiscal and managerial resources away from longer-term innovation-oriented activities (Ji et al., 2024). The asymmetric mediating structure between eastern and non-eastern regions therefore, adds nuance to international debates on smart city heterogeneity, illustrating that similar policy labels can correspond to different combinations of governance modes and innovation-supporting investments.

Finally, the spatial econometric analysis indicates that smart city pilots are associated with positive spillover effects for neighbouring cities. When the spatial weights are redefined using an economic-geographic nested matrix, the spillover effect remains positive. Taking the geographic distribution of innovation into account, it is apparent that this effect is concentrated among cities that are both geographically close and economically similar, while some adjacent but economically weaker neighbours do not experience significant gains. These findings are consistent with research on knowledge flows and absorptive capacity, which emphasises that geographic proximity must be combined with a compatible economic environment and governance capacity for spillovers to materialise (Miguélez & Moreno, 2015a, Miguélez & Moreno, 2015b).

The empirical results also suggest several implications for the design of smart city policies. First, given the stronger average

effects in central and western regions and in small and medium-sized cities, extending smart city-related support to cities with a relatively weak innovation base but adequate governance capacity may provide a useful way to reduce regional gaps, provided that local conditions and fiscal constraints are carefully assessed. Second, the pronounced differences in mediating channels between regions highlight the value of flexible policy packages: in less-developed cities, strengthening basic digital infrastructure and digital government may remain a priority, whereas in more advanced cities, policies that emphasise human capital, collaborative innovation and environmental quality are likely to be more effective. Third, the coexistence of positive spillover effects and persistent innovation divides suggests that regional coordination mechanisms, such as cross-city data platforms or joint innovation projects, can help transform pilot cities into effective demonstration hubs for neighbouring jurisdictions that share a similar economic and governance profile.

Limitations and prospects. Despite the robust findings, this study is subject to several limitations that should be acknowledged.

First, regarding measurement, while the study utilizes robust datasets, reliance on patent-based indicators may not fully capture non-technological innovations—such as improvements in public services and governance—that are central to smart city objectives. This limitation in data coverage may affect the comprehensive assessment of “smart” outcomes and the generalization of the findings.

Second, the binary difference-in-differences design treats pilot status as a uniform intervention, whereas policy implementation differs across regions. Consequently, this empirical design cannot isolate the marginal impact of varying investment intensities or the specific composition of policy bundles, which broad groupings may conceal. Future studies could employ granular data to disentangle the dose-response relationship of policy intensity.

Third, the city-level analysis summarizes average local and spillover effects but cannot uncover the micro-foundations of innovation. The current design identifies macro-channels (TOE) but cannot reveal the detailed mechanisms through which individual firms—the primary agents of innovation—alter their behavioral strategies in response to policy signals.

Finally, these findings are conditioned by China’s state-led institutional context, and mechanisms may operate differently in market-driven models. Further research is warranted to comprehensively understand these micro-dynamics and cross-contextual differences. This understanding will aid in the formulation of integrated smart city strategies and coordinated urban policies, which are essential for fostering broader regional innovation and development.

Data availability

The data used in this study are available from the corresponding author upon reasonable request.

Received: 19 June 2025; Accepted: 30 January 2026;

Published online: 11 February 2026

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Acknowledgements

This research was supported by National Social Science Fund of China (23FJYB009), Soft Science Research Program of Zhejiang Province (2026C15002).

Author contributions

YL and RH wrote the main manuscript text. JZ and RH provided idea, funding, and modifications. JX did the data analysis. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

This article does not contain any studies with human participants performed by any of the authors.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1057/s41599-026-06673-7>.

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